
WEEK 9

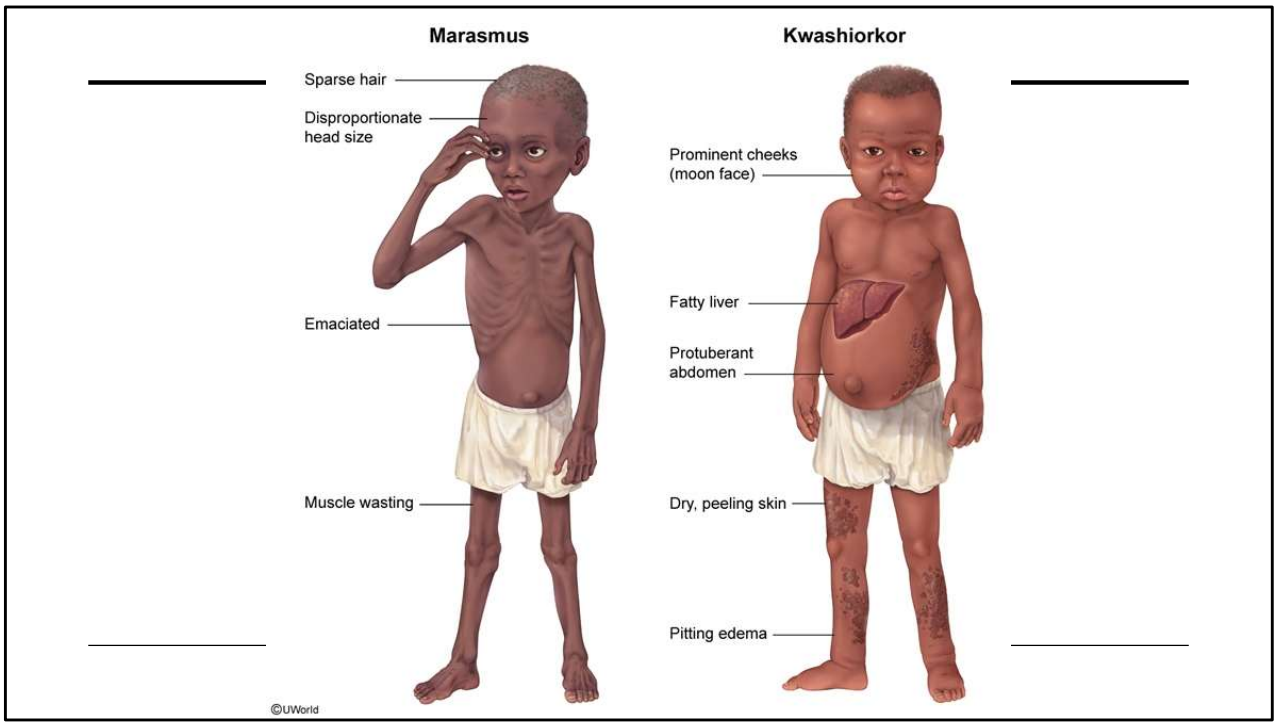
NUTRITION

BMR

BMI

Obesity

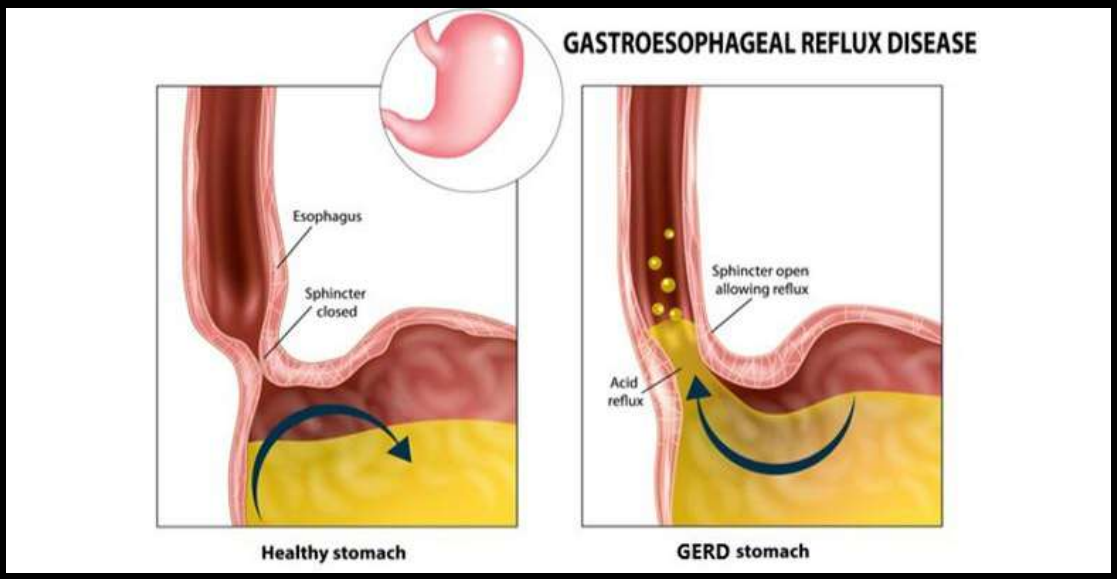
MaInutrition


Marasmus
Kwashiorkor
Sparse hair
Disproportionate head size
Emaciated
Muscle wasting
Prominent cheeks (moon face)
Fatty liver
Protuberant abdomen
Dry, peeling skin
Pitting edema
©UWorld

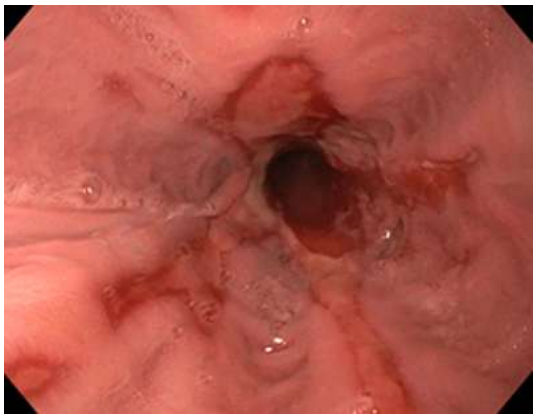
ESOPHAGUS

Dysphagia

GERD


GASTROESOPHAGEAL REFLUX DISEASE
Esophagus
Sphincter closed
Sphincter open allowing reflux
Acid reflux
Healthy stomach
GERD stomach

Esophagitis



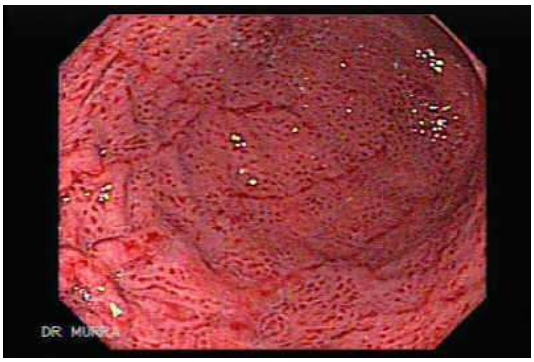
STOMACH

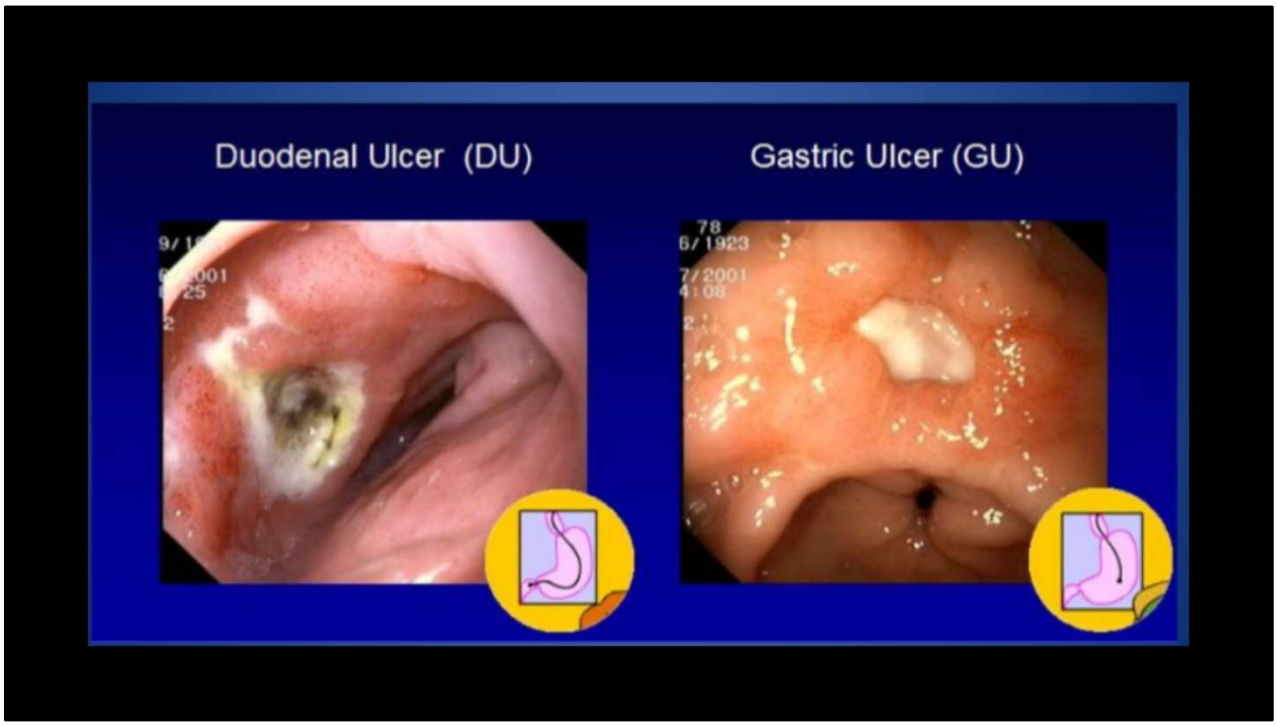
Vomiting

Gastritis

PUD

Gastritis




Duodenal Ulcer (DU)
Gastric Ulcer (GU)

Which of the following is not a potential cause of vomiting?

A. Decreased ICP
B. Chemotherapy
C. Distention of GI tract
D. Inner ear disorders

A

INTESTINES

Constipation & Diarrhea

IBS

IBD – Crohn & UC

Celiac Disease

Gastroenteritis

Appendicitis

Diverticulosis

Colon CA

SBO

A nurse is educating a patient on preventing constipation. What are some risk factors of constipation? Select all that apply.

A. Anxiety
B. Age (older adults)
C. Decreased fluid intake
D. Diet of fruits and vegetables
E. Adequate fiber intake
F. Pregnancy
G. Sedentary lifestyle

A, B, C, F, G

A nurse understands that diarrhea can lead to several complications. Which complication should the nurse address first?

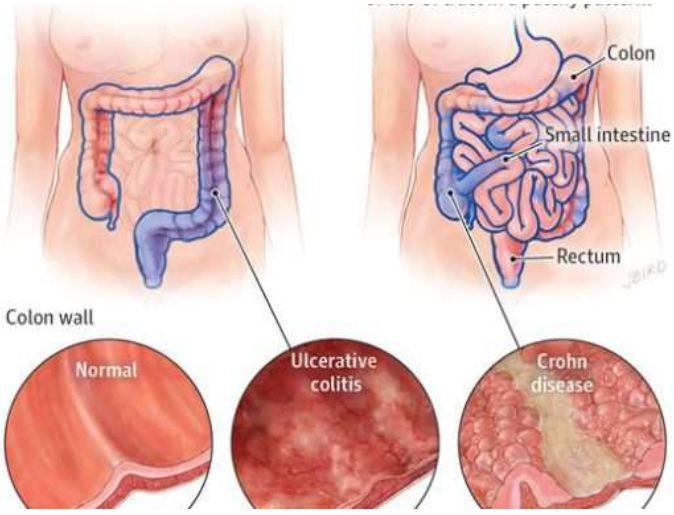
A. Skin irritation and breakdown
B. Fluid and electrolyte imbalance
C. Malnutrition
D. Abdominal pain

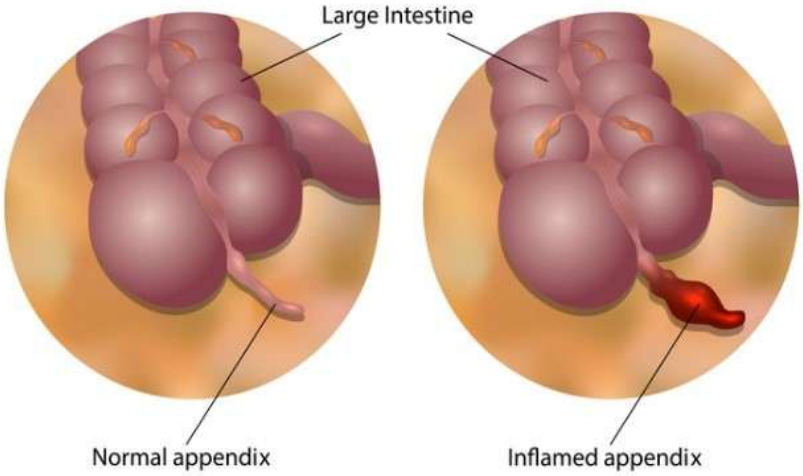
B

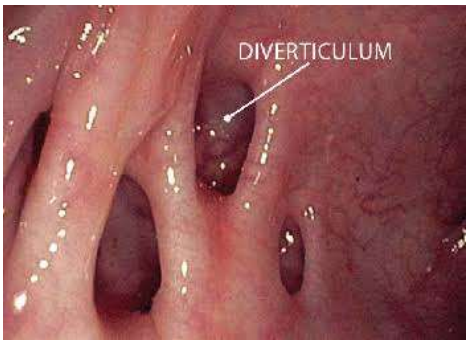
What part of the GI tract absorbs water and electrolytes?

A. Ileum
B. Duodenum
C. Jejunum
D. Colon

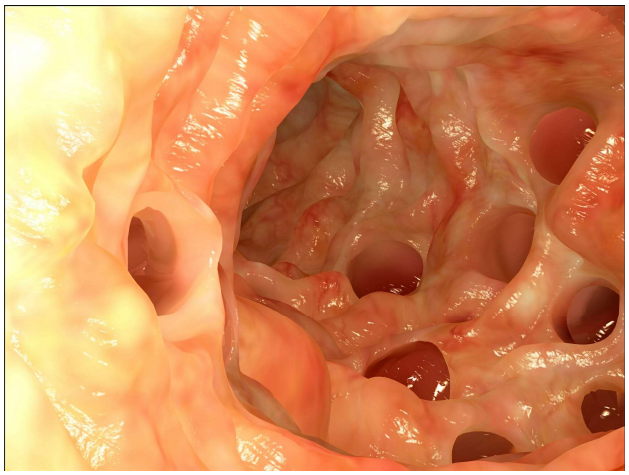
D

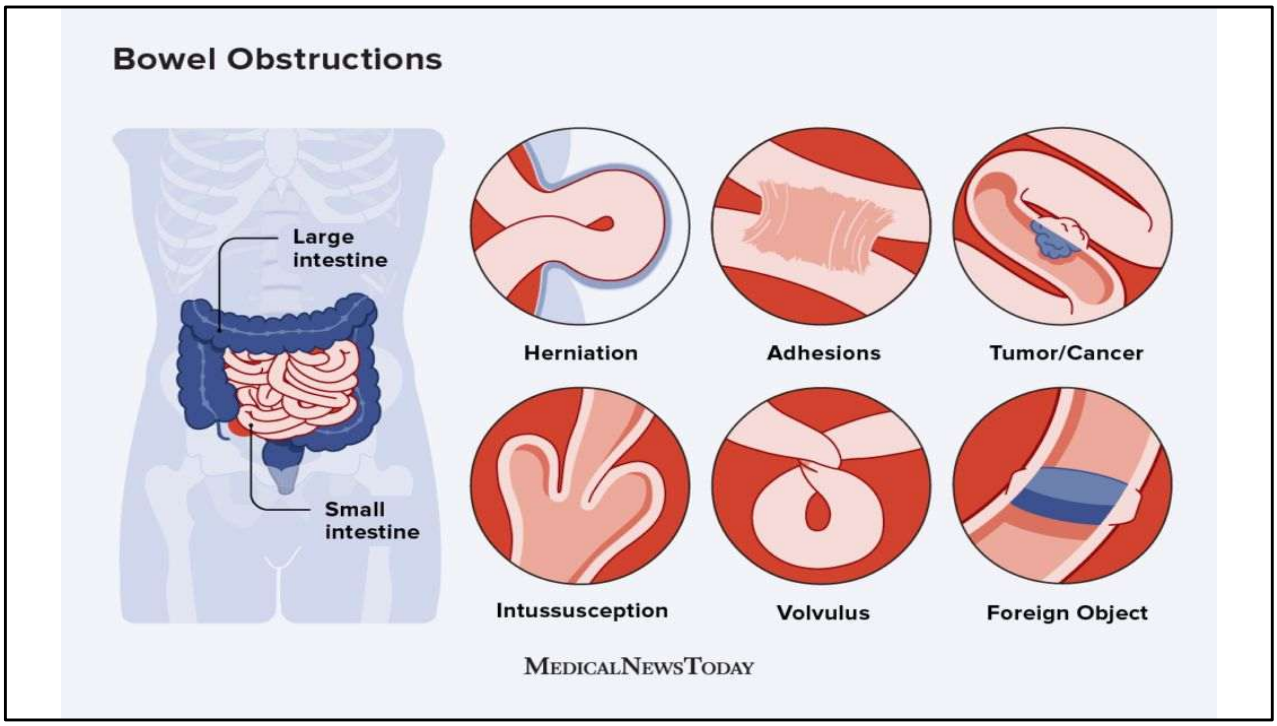

Colon
Small intestine
Rectum
Colon wall
Normal
Ulcerative colitis
Crohn disease


Large Intestine
Normal appendix
Inflamed appendix



Diverticulitis




Bowel Obstructions
Large intestine
Small intestine
Herniation
Adhesions
Tumor/Cancer
Intussusception
Volvulus
Foreign Object
MEDICALNEWSTODAY

Which factor does not contribute to abdominal distention in SBO?

A. Swallowed air

B. Fluid accumulation proximal to blockage

C. Vomiting

D. Intestinal venous compression

C

A nurse is assessing a patient in the ER. What symptoms would the nurse recognize as a potential bowel obstruction? Select all that apply.

A. Abdominal pain
B. Hypoactive or hyperactive bowel sounds
C. Positive Murphy's sign
D. Vomiting, nausea, loss of appetite
E. Regular, soft stools
F. Distended abdomen

A, B, D, F

LIVER

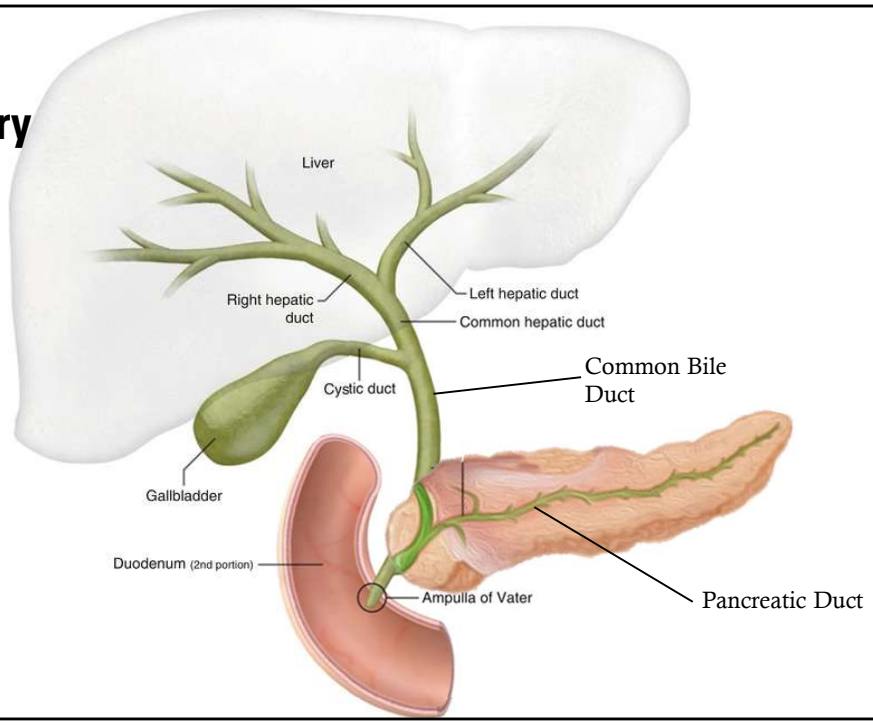
Jaundice

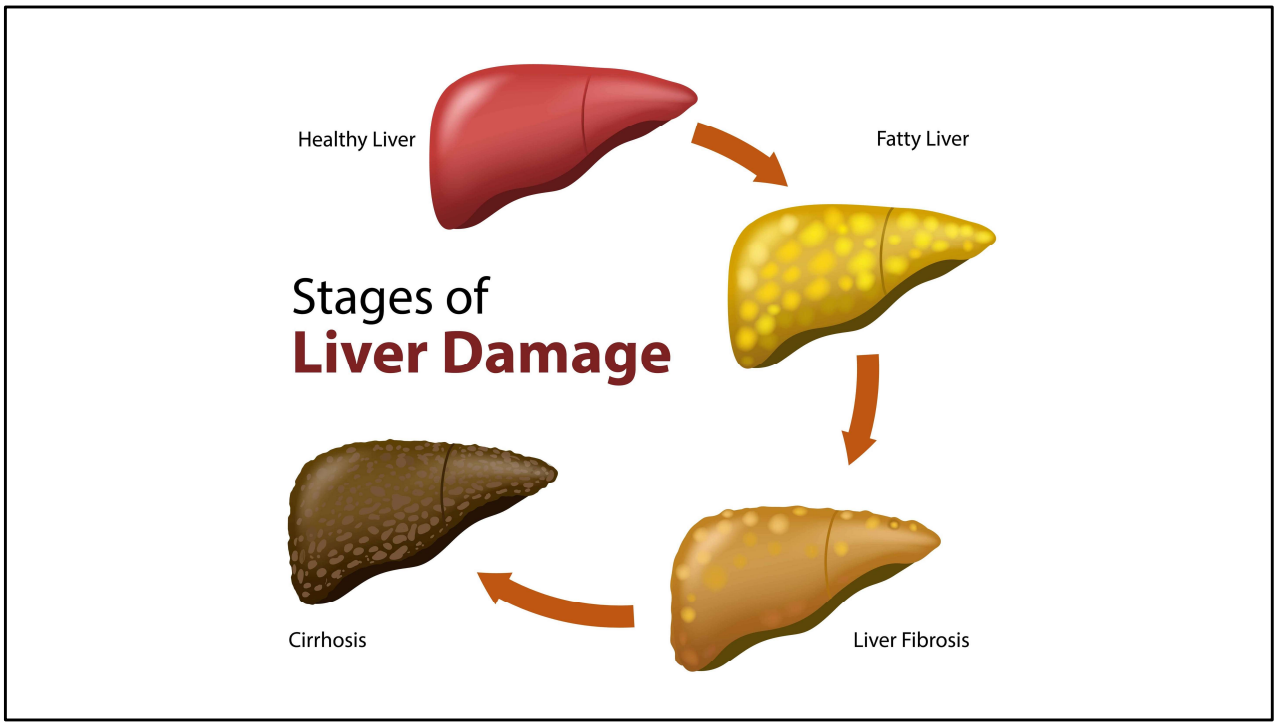
Hepatitis

Cirrhosis

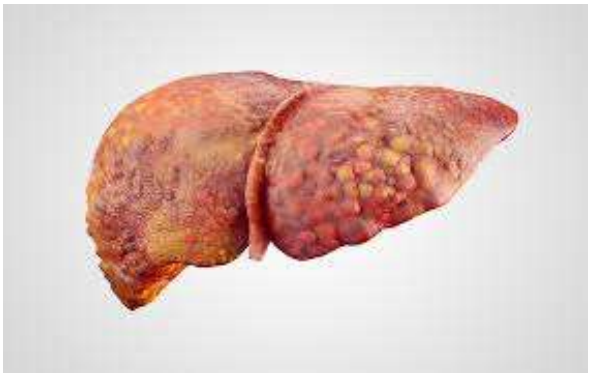
Liver CA

Hepatobiliary System

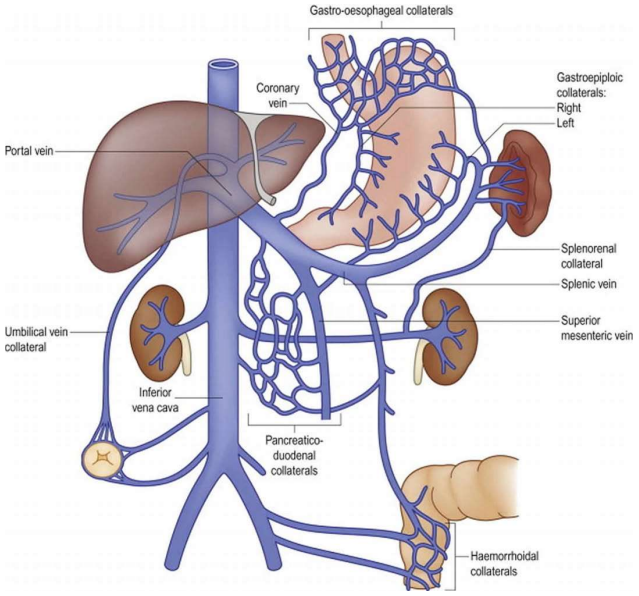



Healthy Liver
Fatty Liver
Stages of
Liver Damage
Cirrhosis
Liver Fibrosis

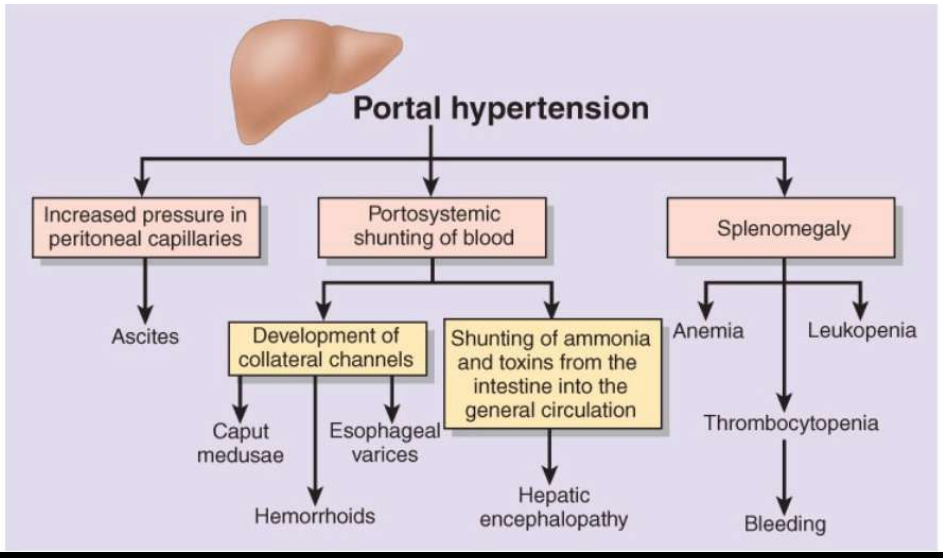




Cirrhosis



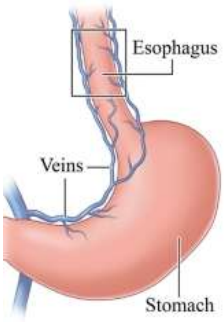
Portal System

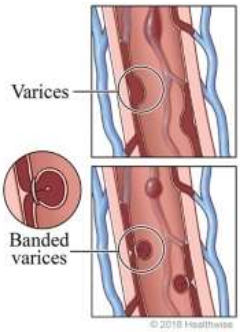

Portal hypertension
Increased pressure in peritoneal capillaries
Portosystemic shunting of blood
Splenomegaly
Ascites
Development of collateral channels
Shunting of ammonia and toxins from the intestine into the general circulation
Anemia
Leukopenia
Caput medusae
Esophageal varices
Thrombocytopenia
Hemorrhoids
Hepatic encephalopathy
Bleeding

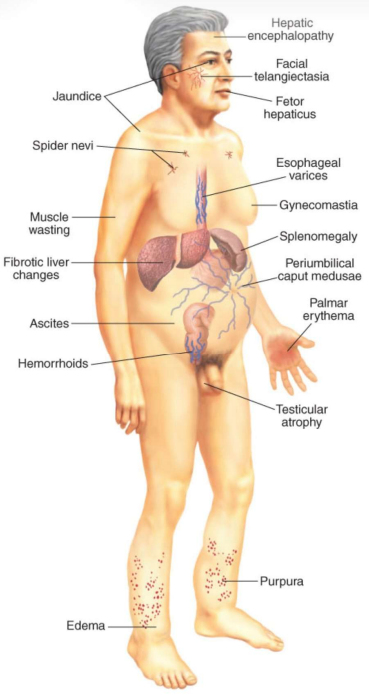


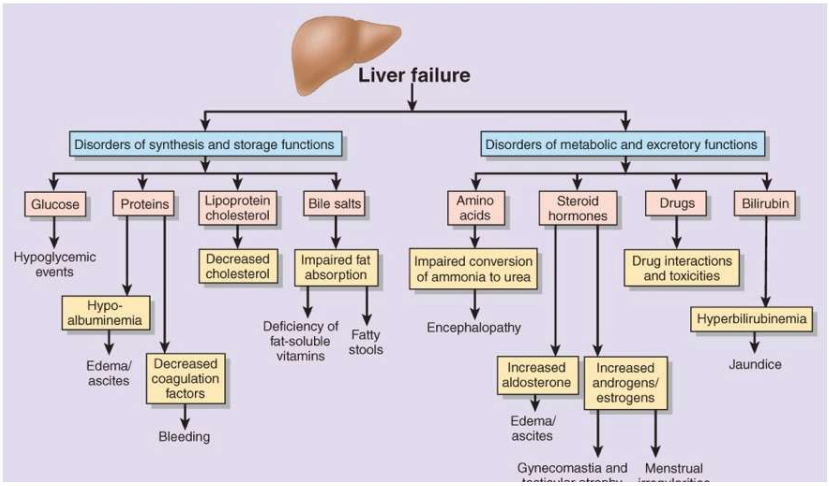
ESOPHAGEAL VARICES








Hepatic encephalopathy
Facial telangiectasia
Jaundice
Fetor hepaticus
Spider nevi
Esophageal varices
Gynecomastia
Muscle wasting
Splenomegaly
Fibrotic liver changes
Periumbilical caput medusae
Palmar erythema
Ascites
Hemorrhoids
Testicular atrophy
Purpura
Edema


Liver failure
Disorders of synthesis and storage functions
Disorders of metabolic and excretory functions
Glucose
Proteins
Lipoprotein cholesterol
Bile salts
Amino acids
Steroid hormones
Drugs
Bilirubin
Hypoglycemic events
Hypo-albuminemia
Edema/ ascites
Decreased coagulation factors
Bleeding
Decreased cholesterol
Impaired fat absorption
Deficiency of fat-soluble vitamins
Fatty stools
Impaired conversion of ammonia to urea
Encephalopathy
Increased aldosterone
Edema/ ascites
Increased androgens/ estrogens
Gynecomastia and
Menstrual
Drug interactions and toxicities
Hyperbilirubinemia
Jaundice

The nurse cares for a client with hepatitis that has progressed to cirrhosis. Which lab result is the nurse likely to observe from this client?

A. Decreased platelet count
B. Elevated serum sodium
C. Elevated serum albumin
D. Decreased alanine aminotransferase

A

A client with cirrhosis has increasing abdominal girth. Which laboratory result does the nurse prioritize examining prior to notifying the healthcare provider?

A. Potassium
B. Liver enzymes
C. Albumin
D. Sodium

C

The nurse reviews the laboratory results for a patient with cirrhosis. The nurse expects which likely laboratory result?

A. The client's international normalized ratio is decreased
B. The client's serum vitamin K level is lower than normal
C. The client's serum vitamin D level is higher than normal
D. The client's prothrombin time is lower than normal

B

A nurse assesses a client with end-stage liver failure. The nurse recognizes which symptom of end-stage liver failure in this client?

A. Spontaneous bruising
B. Increased activity level
C. Blue-tinged extremities
D. Increased urine output

A

Which type of hepatitis is transmitted by the fecal-oral route and is a short-term infection?

A. Hepatitis A
B. Hepatitis B
C. Hepatitis C
D. Hepatitis D

A

What causes ascites in the cirrhotic patient?

A. Decreased vascular oncotic pressure from low blood albumin levels
B. Increased arterial hydrostatic pressure from hypertension
C. Increased sodium in the interstitial space from increased peritoneal permeability
D. Decreased venous hydrostatic pressure from portal hypertension

A

What are possible signs and symptoms of cirrhosis? Select all that apply.

A. Ecchymoses
B. Asterixis
C. Confusion
D. Hypertension
E. Pallor
F. Melena
G. Pruritus

A, B, C, F, G

Which types of hepatitis increase the risk of developing liver cancer? Select all that apply.

A. Hepatitis A
B. Hepatitis B
C. Hepatitis C
D. Hepatitis D

B, C

Which statement is true of the icteric stage of hepatitis?

A. The icteric stage involves fatigue, anorexia, nausea, and increased liver function tests (LFTs)

B. The icteric stage is when the patient has jaundice, dark amber urine, and light colored stools

C. The icteric stage usually lasts about 8-16 weeks

D. The icteric stage is only present in hepatitis A

B

Which of the following lab values would be expected in a patient with cirrhosis? Select all that apply.

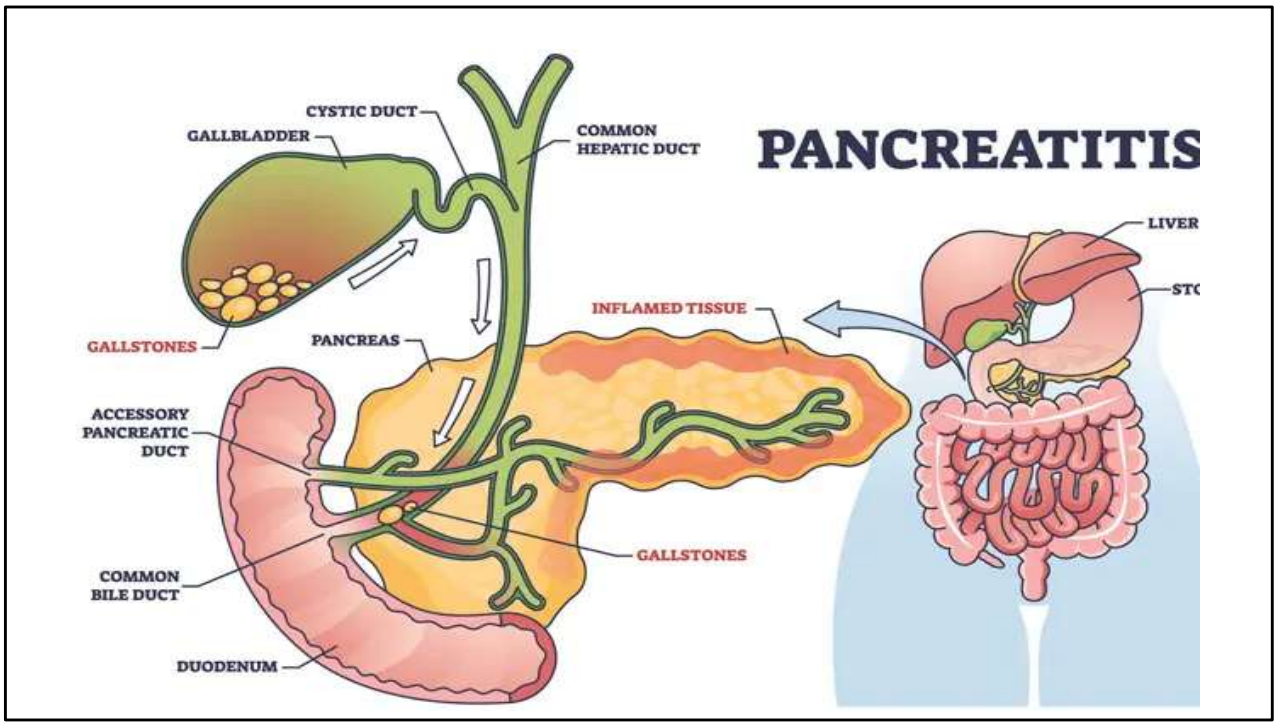
A. Increased platelets
B. Decreased INR
C. Increased bilirubin
D. Decreased ALT, AST, alkaline phosphatase
E. Increased albumin
F. Decreased hemoglobin

C, F

cirrhosis impairs clotting so there would be low platelets and high INR. High bilirubin causes jaundice. There is low albumin due to impaired protein synthesis in the liver. Liver enzymes (ALT, AST, alk phos) will be high due to lots of damage → enzymes released from damaged liver cells

PANCREAS

Pancreatitis


PANCREATITIS
CYSTIC DUCT
GALLBLADDER
COMMON HEPATIC DUCT
LIVER
INFLAMED TISSUE
GALLSTONES
PANCREAS
ACCESSORY PANCREATIC DUCT
GALLSTONES
COMMON BILE DUCT
DUODENUM

The nurse assesses a client admitted with acute pancreatitis. Which signs and symptoms is the nurse likely to observe? Select all that apply.

A. Tachycardia
B. Back pain
C. Hyperglycemia
D. Hypercalcemia
E. Hematochezia

A, B, C

What are the most common causes of pancreatitis?

A. Gallstones and alcoholism

B. Kidney stones

C. Peritonitis and appendicitis

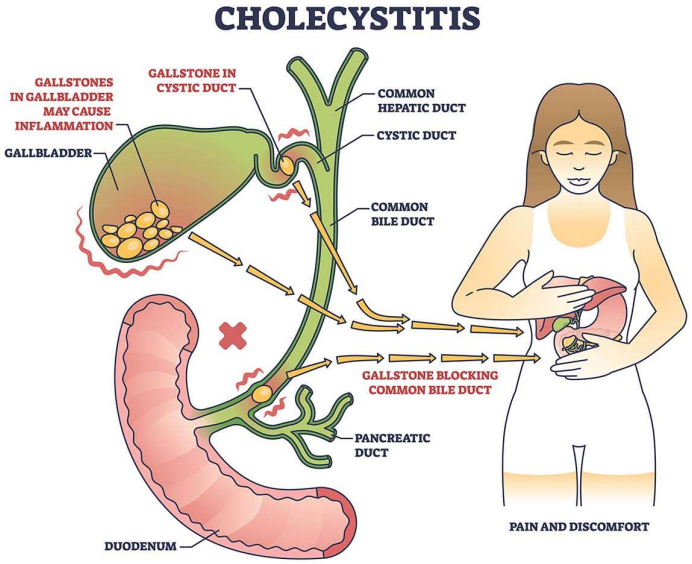
D. Gastritis and gastric resection

A

GALLBLADDER

Cholelithiasis

Cholecystitis


CHOLECYSTITIS
GALLSTONES IN GALLBLADDER MAY CAUSE INFLAMMATION
GALLSTONE IN CYSTIC DUCT
COMMON HEPATIC DUCT
CYSTIC DUCT
GALLBLADDER
COMMON BILE DUCT
GALLSTONE BLOCKING COMMON BILE DUCT
PANCREATIC DUCT
DUODENUM
PAIN AND DISCOMFORT



CHOLELITHIASIS

An ER nurse cares for a patient who reports abdominal pain and vomiting. For each sign and symptom, select to specify whether the signs and symptoms may indicate a diagnosis of cholecystitis, pancreatitis, or both.

Signs and Symptoms	Cholecystitis	Pancreatitis	Both
WBC 15,000			
Temperature 101.3F			
Sensitivity to fatty foods			
Elevated lipase			
Right upper quadrant pain			
Epigastric pain radiating to the back			
Imaging reveals gallstones			

1. Both
2. Both
3. Chole
4. Pancreatitis
5. Chole
6. Both
7. Both

The nurse cares for a patient admitted with acute cholecystitis. The nurse identifies which risk factor associated with gallstone formation and cholecystitis when performing the admission assessment?

A. The client has never been pregnant or had children
B. The client has a history of appendicitis
C. The client is taking oral contraceptives
D. The client's ethnic group is African American

C